

Embodiment in Action: Embodied Experiences of More-Than-Human Entanglements

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Abstract

This meet-up brings together HCI researchers, designers, and practitioners to explore embodiment and more-than-human relations through a shared, situated, and embodied experience. While fourth-wave HCI has increasingly addressed the entanglement of bodies, technologies, and systems of power, such discussions are often abstract and disembodied. This session counters that trend by centring participants' lived, bodily states, such as hunger, fatigue, joy, and discomfort, through a series of rotating, themed stations inspired by

embodied design and World Café methods. Participants will reflect on how social inequalities, technologies, and non-human actors are inscribed in and experienced through the body. By approaching embodiment with our own bodies at the centre, this meet-up aims to cultivate a sense of community, inspire new research perspectives, and foster long-term collaborations. Attendees will leave not only with insights but with a felt sense of relationality that extends beyond the session.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; *HCI design and evaluation methods*.

Keywords

embodiment, feminist HCI, fourth-wave HCI, more-than-human

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1 Introduction

As researchers, designers, and users of technology, we are entangled with a plethora of relations, both human and non-human, and are situated within power structures that inscribe themselves into our bodies. These Social inequalities are both written into bodies and influence the way others perceive them, embodying social relationships. More-than-human actors, such as microbes and bacteria, the plants and animals we eat, the surrounding nature, and technical artefacts in, on, and around us, are also part of our corporeality. Fourth-wave HCI research draws attention to our more-than-human entanglements [1] and the mutual shaping of technology and society [10], criticising essentialised notions of users, and gendered, racialised, and classed normative expectations inscribed into technologies and bodies [3, 10].

Paradoxically, the more we aim to intellectually grasp our corporeality and our human and non-human relations, the further we drift away from an embodied, felt, and sensed towards a rationalised understanding, thus removing ourselves from the very topic of our inquiry and separating ourselves from our bodies, study subjects, and relations. This separation into self and others is the cause of many social injustices, and has already been addressed methodologically by approaches such as autobiographical design, autoethnography, and bodystorming [18].

Addressing these challenges, this meet-up aims to facilitate discussions of our corporeal reality within sociotechnical systems of oppression and entangled with more-than-human actors to develop novel directions in HCI. As such, we invite HCI researchers, designers, and practitioners who are interested in and/or work with notions of embodiment and more-than-human perspectives on a theoretical, methodological, or design level. Our goal is to create a sense of community, belonging, and sharedness in our collective goals to actively question and avoid inscribing essentialised notions into technologies, and carefully consider the harms of designs that naturalise systems of oppression [10].

2 Structure and Activities

The meet-up is more than just a rational exchange with others – it foregrounds the participants' bodies and the spaces they occupy. As such, it manifests, reinforces, or reminds us of the importance of what we aim to collectively experience, investigate, and express in research. By grounding our approach in bodily experiences, we invite discussions on embodiment and more-than-human perspectives, premised on our vulnerable, situated corporeality, and corporeal sharedness and connectedness. We will explore and reimagine notions of embodiment and more-than-human perspectives to inspire novel perspectives on methodology, theory, and design. Our goal is to assemble a global tribe – before they disappear into the conference, leaving with a feeling of shared understanding, connectedness, and community lasting beyond the 90-minute meet-up. Long term, the meet-up aims to queer, quiz, and question dominant notions of embodiment in the CHI community and affirm one's

own embodiment and vulnerabilities as researchers as a valuable source for research.

Participants will embody the conversations during the meet-up and carry them with them, connecting on more than an academic wavelength. By disrupting the trend of intellectualisation within conferences, we highlight the shared, often overlooked realities of our bodies – we are tired, we go to the toilet, we eat the bodies of other things, which we then embody, and we are connected to other bodies.

Introduction & Grounding (10–15 min). After a welcome and introduction, a grounding exercise will centre participants in their embodied experience of 'being-at-CHI' – such as tiredness from jet lag that manifests our roots to home, a dry throat due to dehydration, or feeling a proud grin stretch across one's face after a successful paper presentation – to set the tone for the meet-up.

Themed Stations – Embodied Reflections (60 min). Five themed stations will invite participants to connect to the bodily experiences related to 'being-at-CHI', inspired by embodied design methods (e.g., [5, 9, 18]) and world cafés [6]. To encourage cross-pollination and building-on other attendees' prior reflections, participants will move intermittently between stations, encountering embodied prompts such as: *In this moment, what are you experiencing in your body? Please share, move, draw, or craft a facet of your experience you feel comfortable sharing. How does your state of being connect with other past and present times, or other places? What relations to others are inscribed in our bodies? How do we interpret signals from the body, and how is this facilitated, hindered, or supported by technologies (e.g., trackers or online content)? How can we make connecting to our state of being embodied an act of resistance?* Overall, each participant will pollinate 2-4 stations. The stations include (i) *Hunger & Thirst*, (ii) *Tiredness & Rest*, (iii) *Joy & Excitement*, (iv) *Discomfort, Pain, Impairments*, (v) *Social Connectedness*. At each station, a member of the organiser team will facilitate the embodied reflection and track overall topics of discussion to identify themes for future meet-ups.

Relationality Exercise & Closing (5–10 min). We will close with a joint embodied activity, where participants mirror each other's movements in a dance of relationality, fostering a deep sense of connection. The meet-up will conclude with closing remarks including voluntary reflections of the attendees as well as envisioning meaningful next steps together.

3 Organisers

We bring together an international team of organisers with different research focuses and backgrounds to ensure attracting a diverse audience and thus a range of perspectives. We approach embodied research in HCI through different lenses – from the theoretical and methodological to the designerly and practical. Our research spans a set of interrelated fields of interest, including more-than-human design [1], embodied empowerment [14], body comportment in the design of embodied interactions [15], embodied and relational understandings of digital agency [2], existential HCI [11], embodied and collaborative sensemaking [16, 17], diversity in HCI [8] and participatory design [4], defining new design [13] as well as evaluation [7, 12] approaches. Across these intersecting topics, our

research shares common through lines, such as engagement with more-than-human, feminist, and critical perspectives.

4 Post Meet-Up Plans

This meet-up is intended to kick-start a longer-term community of shared interest and inspiration. As such, we envision follow-up meet-ups in different formats, settings, and contexts. To facilitate long-term community building, we will invite participants to sign up for a mailing list and/or a Discord channel to create a knowledge space for sharing thoughts and ideas and foster collaboration. We aim to arrange similar and/or informal meetings at other HCI conferences, such as ACM Designing Interactive Systems Conference (DIS), ACM Participatory Design Conference (PDC), and ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW). Thus, we raise awareness for the topic, fostering going beyond ‘productive’ conventions towards queering, quizzing and questioning what our bodies are carrying, and allow thinking from our own bodies and experiences.

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References

- [1] Yoko Akama, Ann Light, and Takahito Kamiyama. 2020. Expanding Participation to Design with More-Than-Human Concerns. In *Proceedings of the 16th Participatory Design Conference 2020 - Participation(s) Otherwise - Volume 1* (Manizales, Colombia) (PDC '20). Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3385010.3385016
- [2] Diotima Bertel, Julia Himmelsbach, Stephanie Schwarz, David Sellitsch, Andreas Wildmann, Alexander Schmölz, and Manfred Tscheligi. 2025. Play It Till You Make It: The Potential of Playful Role Enactment to Foster Digital Agency. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 325, 18 pages. doi:10.1145/3706598.3713258
- [3] Daniel A. Epstein, Nicole B. Lee, Jennifer H. Kang, Elena Agapie, Jessica Schroeder, Laura R. Pina, James Fogarty, Julie A. Kientz, and Sean Munson. 2017. Examining Menstrual Tracking to Inform the Design of Personal Informatics Tools. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 6876–6888. doi:10.1145/3025453.3025635
- [4] Jeanette Falk, Anna Blumenkranz, Moritz Kubesch, Ralf Vetter, Lisa Hofer, and Christopher Frauenberger. 2024. Designing Diverse Pathways for Participation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 508, 16 pages. doi:10.1145/3613904.3642240
- [5] Sarah Fdili Alaoui, Thecla Schiphorst, Shannon Cuykendall, Kristin Carlson, Karen Studd, and Karen Bradley. 2015. Strategies for Embodied Design: The Value and Challenges of Observing Movement. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition* (Glasgow, United Kingdom) (C&C '15). Association for Computing Machinery, New York, NY, USA, 121–130. doi:10.1145/2757226.2757238
- [6] Christa Fouché and Glenda Light. 2011. An Invitation to Dialogue: ‘The World Café’ In Social Work Research. *Qualitative Social Work* 10, 1 (2011), 28–48. doi:10.1177/1473325010376016 arXiv:https://doi.org/10.1177/1473325010376016
- [7] Cornelia Gerdenitsch, Matthias Weinhofer, Jaison Puthenkalam, and Simone Kriglstein. 2022. Upward Influence Tactics: Playful Virtual Reality Approach for Analysing Human Multi-robot Interaction. In *Entertainment Computing – ICEC 2022: 21st IFIP TC 14 International Conference, ICEC 2022, Bremen, Germany, November 1–3, 2022, Proceedings* (Bremen, Germany). Springer-Verlag, Berlin, Heidelberg, 76–88. doi:10.1007/978-3-031-20212-4_6
- [8] Julia Himmelsbach, Stephanie Schwarz, Cornelia Gerdenitsch, Beatrix Wais-Zechmann, Jan Bobeth, and Manfred Tscheligi. 2019. Do We Care About Diversity in Human Computer Interaction: A Comprehensive Content Analysis on Diversity Dimensions in Research. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3290605.3300720
- [9] Kristina Höök, Sara Eriksson, Marie Louise Juul Søndergaard, Marianela Ciolfi Felice, Nadia Campo Woytuk, Ozgun Kilic Afsar, Vasiliki Tsaknaki, and Anna Ståhl. 2019. Soma design and politics of the body. In *Proceedings of the Halfway to the Future Symposium 2019*. 1–8.
- [10] Os Keyes, Burren Peil, Rua M. Williams, and Katta Spiel. 2020. Reimagining (Women’s) Health: HCI, Gender and Essentialised Embodiment. *ACM Trans. Comput.-Hum. Interact.* 27, 4, Article 25 (Aug. 2020), 42 pages. doi:10.1145/3404218
- [11] Quynh Nguyen, Julia Himmelsbach, Diotima Bertel, Olivia Zechner, and Manfred Tscheligi. 2022. What Is Meaningful Human-Computer Interaction? Understanding Freedom, Responsibility, and Noos in HCI Based on Viktor Frankl’s Existential Philosophy. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference* (Virtual Event, Australia) (DIS '22). Association for Computing Machinery, New York, NY, USA, 654–665. doi:10.1145/3532106.3533484
- [12] Quynh Nguyen, Emma Jaspaert, and Birgit Harthum. 2024. Talk the Talk, Walk the Walk: A Case Study on Systematic Evaluation in Participatory Design. In *Proceedings of the Participatory Design Conference 2024: Full Papers - Volume 1* (Sibu, Malaysia) (PDC '24). Association for Computing Machinery, New York, NY, USA, 133–144. doi:10.1145/3666094.3666103
- [13] Quynh Nguyen, Emma Jaspaert, Markus Murtinger, Simone Kriglstein, and Manfred Tscheligi. 2024. CoLEBricks: Co-Designing Virtual Reality Scenarios with Generative Building Blocks. In *Proceedings of the 2024 International Conference on Advanced Visual Interfaces* (Arenzano, Genoa, Italy) (AVI '24). Association for Computing Machinery, New York, NY, USA, Article 52, 5 pages. doi:10.1145/3656650.3656686
- [14] Bojana Nikolovska, Maria Normark, and Helga Sadowski. 2025. Imagining with the Body: Speculative Designs for Women’s Embodied Empowerment in Feminist Self-Defense. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 406, 13 pages. doi:10.1145/3706598.3713975
- [15] Dorothé Smit, Nathalia Campreguer França, and Verena Fuchsberger. 2025. Interacting Like a Girl: Considering Body Comportment in the Design of Embodied Interactions. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference* (DIS '25). Association for Computing Machinery, New York, NY, USA, 2950–2958. doi:10.1145/3715336.3735800
- [16] Dorothé Smit, Bart Hengeveld, Martin Murer, and Manfred Tscheligi. 2022. Hybrid Design Tools for Participatory, Embodied Sensemaking: An Applied Framework. In *Proceedings of the Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Daejeon, Republic of Korea) (TEI '22). Association for Computing Machinery, New York, NY, USA, Article 26, 10 pages. doi:10.1145/3490149.3501332
- [17] Dorothé Smit, Ruth Neubauer, and Verena Fuchsberger. 2021. Distributed Collaborative Sensemaking: Tracing a Gradual Process. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Salzburg, Austria) (TEI '21). Association for Computing Machinery, New York, NY, USA, Article 103, 12 pages. doi:10.1145/3430524.3446079
- [18] Danielle Wilde, Anna Vallgård, and Oscar Tomico. 2017. Embodied Design Ideation Methods: Analysing the Power of Estrangement. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 5158–5170. doi:10.1145/3025453.3025873